



**Team
AXIOMATA**

PROJECT OVERVIEW AND TEAM PROFILE

Developed By
Team AXIOMATA

Developed for
Waypoint Group (Pvt) Ltd.

Access to RELAY

1. About Axiomata

Axiomata is a competitive software team participating in the Rootcode Tech-Triathlon 2026 hackathon. The team combines expertise in software engineering, machine learning, UX design and logistics optimisation. Our entry product, **Relay**, was conceived, designed and built end-to-end during the competition cycle.

2. The Problem We Solved

Waypoint delivers fast-moving consumer goods to 120 retail outlets across Sri Lanka from two depots, Peliyagoda and Kandy. Every afternoon at 4 PM the stores' orders close and a dispatcher must fit the next day's demand into the vehicles that are actually available, subject to a web of hard rules:

- Reefer trucks required for chilled goods
- Van-only outlets that trucks cannot reach
- One brand and one district per trip
- A 270-minute pre-dawn window for Fresh goods (03:30–08:00)
- Weekly fuel quotas per vehicle
- Vehicles in the workshop on any given day
- Delivery-window constraints at mall and street outlets

Demand routinely exceeds cold capacity, so some orders must be deferred - and the same outlets keep getting skipped. When the plan meets reality, fresh problems arise: a short shipment at the dock, a van losing signal in the hill country, a damaged case at the store. Relay addresses every stage of that day end-to-end.

3. Product Vision - One Plan, Four Faces

Relay is built on a single unifying concept: one optimised plan, presented differently to each person who carries it out. Rather than four disconnected apps, every role works on the same live data; a change made anywhere reaches everyone it affects within seconds.

Role	Device	Key Capabilities
Dispatcher	Desktop (1440 px+)	CP-SAT planning, deferral lever, drag-and-drop order moves, dark-van tracking, shortfall decisions
Loader	Tablet / phone	Reverse-stop-order load list, line ticking, flag shortfalls, swipe-to-release with seal & temperature
Driver	Phone (PWA, offline)	Stop-by-stop run, arrival swipe, photo + signature proof, offline outbox, Sinhala/Tamil/English
Store Manager	Desktop or phone	Order placement, real-time delivery tracking, receipt confirmation with per-line issues

4. Technology Stack

Layer	Technology
Frontend	React 19 · TypeScript · Vite · React Query · Dexie (IndexedDB) · PWA
Backend API	FastAPI · SQLAlchemy 2 · psycpg 3 · Alembic · Pydantic 2 · Python 3.12
Planning Engine	OR-Tools CP-SAT (Google) - pure Python package
ML / Prediction	LightGBM with heuristic fallback
Database	PostgreSQL 16 (Cloud SQL on GCP / local Docker)
Live Updates	Postgres LISTEN/NOTIFY → Server-Sent Events
Offline Sync	IndexedDB outbox (Dexie) with idempotent server-side apply
Infrastructure	Cloud Run · Cloud SQL · Cloud Build · Artifact Registry · Secret Manager
CI/CD	GitHub Actions · Docker Compose · Playwright · pytest

5. Key Metrics

Metric	Value
Outlets served	120 across 12 districts
Vehicles	60 (trucks and vans, ambient and reefer)
Depots	2 (Peliyagoda, Kandy)
Orders planned (demo)	~137 Wednesday orders + carry-overs
Typical solve time	< 10 seconds (2 vCPU)
Test coverage	30+ Python API tests, Playwright 4-role e2e walkthrough
Languages supported	English · Sinhala · Tamil (driver app)
Offline capability	Full driver and dock operation without connectivity